

Phase 1: Setup & Foundations

1. Create project folder & virtual environment

- `mkdir timesheet_project && cd timesheet_project`
- `python -m venv venv`
- Activate environment and install: `django pandas plotly openpyxl`

2. Start Django project & app

- `django-admin startproject timesheet_project .`
- `python manage.py startapp timesheet_app`

3. Configure settings

- Add `timesheet_app` to `INSTALLED_APPS`
- Set up `templates/` and `static/` directories

Phase 2: Authentication

4. Enable Django's built-in login system

- Create `login.html` and `logout.html`
- Use `User` model for username/password
- Protect dashboard routes with `@login_required`

Phase 3: Data Handling

5. Set up Excel data source

- Place Excel file in `data/timesheets.xlsx`
- Write Pandas function to load data automatically at startup

6. Filtering logic

- Functions to filter by:
 - Date range (May–August, or all)
 - Employee dropdown (individual selection)
- Return filtered `DataFrame`

Phase 4: Visualization

7. Create plotting functions

- Use `Plotly` for interactive charts

- Example: bar chart (billable hours), line chart (estimated hours)

8. Integrate plots into Django views

- Render charts in dashboard.html
- Use plotly.offline.plot or django-plotly-dash

Phase 5: Frontend

9. Build templates

- login.html → login form
- dashboard.html → filters + chart display
- plot_view.html → individual employee timesheet

10. Add styling

- Basic CSS in static/style.css
- Keep clean, responsive layout

Phase 6: Testing & Iteration

11. Run locally

- python manage.py runserver
- Test login, filtering, and chart rendering

12. Fix bugs & refine

- Ensure Excel loads automatically
- Validate filters work correctly
- Improve chart interactivity

Phase 7: Future Extensions

13. Browser/intranet deployment

- Later: deploy with Gunicorn + Nginx on internal server
- Add HTTPS if needed

14. Enhancements

- Role-based access (managers vs employees)
- Export filtered reports (Excel/PDF)
- Store data in SQLite/PostgreSQL instead of Excel

